

Data models in Power BI

Relationships and cardinality in Power BI

Defining relationships in Power BI

As in relational databases, relationships are essential in Power BI for **connecting and associating data from different tables** in a data model.

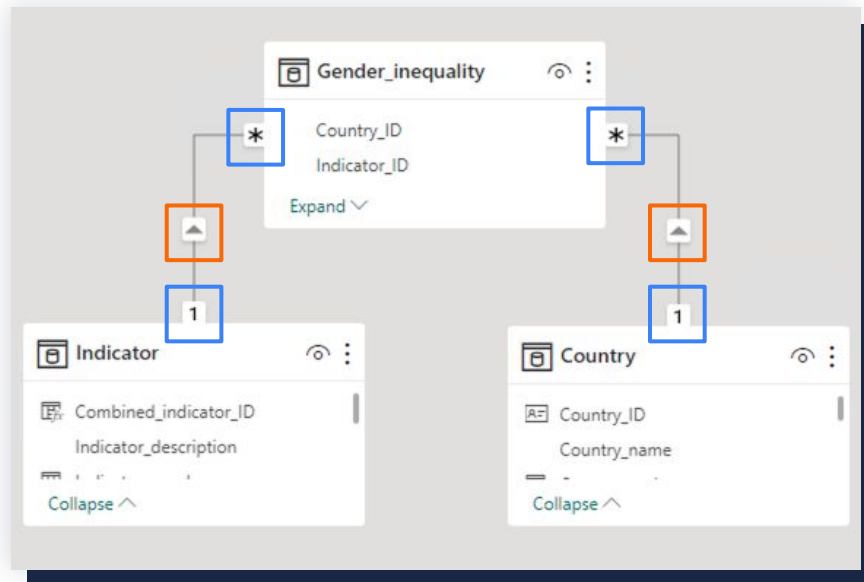
When we define relationships in Power BI, we need to consider the following:

Cardinality

The **number of related records** in one table that can be linked to a single record in another table.

Directionality

Defines **how filtering** in one table affects another table.

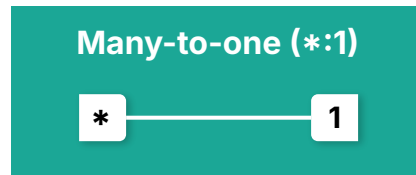
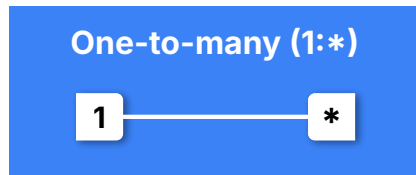
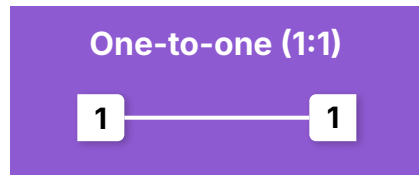


Cardinality

We can have the same type of relationships in Power BI that we would normally have in a relational database, including **one-to-one**, **one-to-many**, **many-to-one**, and **many-to-many relationships**.

“Cardinality” is a mathematical term that refers to the **number of elements in a given set**. Similarly, in a database it represents the **number of related records in one table** that can be **linked to a single record** in another table.

The **four main types of cardinality** in Power BI are:



We can have all of these **cardinalities in a single data model**. However, **one-to-many cardinalities are more common** in structured data models such as the star schema.



Directionality

Unlike SQL, Power BI introduces the concept of "cross-filter direction", also known as directionality, which **defines how filtering in one table affects another table**.

Single

Filtering in the **"One" table** (the table with a single record on one side of the relationship) **impacts the "Many" table**, allowing us to filter related records.

This is often called a **single relationship**.

Both

Filtering in **both tables** affects each other, enabling two-way filtering. Although useful, it introduces complexity.

This is often called **bi-directional relationships**. In some cases, we **may want both tables to impact each other's filtering**. **One-to-one** cardinalities can only ever be bi-directional relationships.

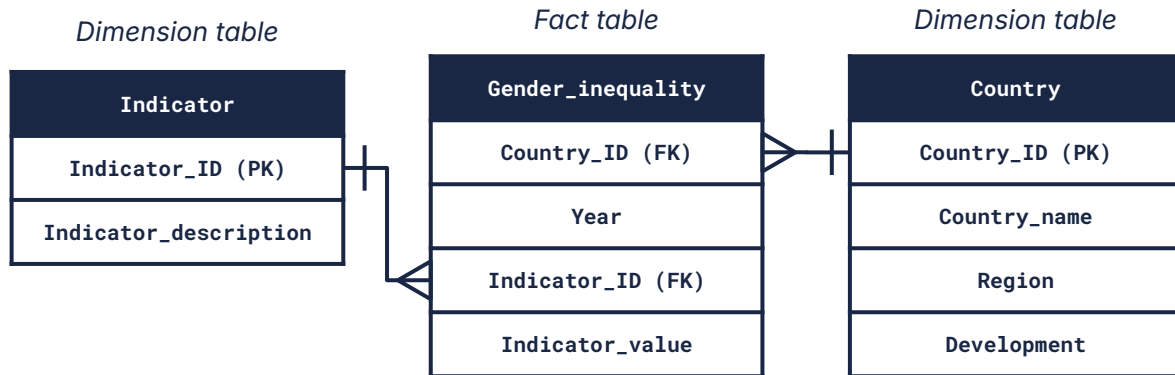
Many-to-many, **one-to-many**, and **many-to-one** cardinalities can **either be single or bi-directional**.

Dataset example

UN SDG 5: Gender equality

This database hosts gender-disaggregated data and statistics covering **geographic location**, **education**, **health**, access to **economic opportunities**, public life and **decision-making**, and **agency**.

We want to look at **gender parity (or equality)** and inequality across different geographical areas. For a simple star schema data model, we simply break up the dataset into three tables:



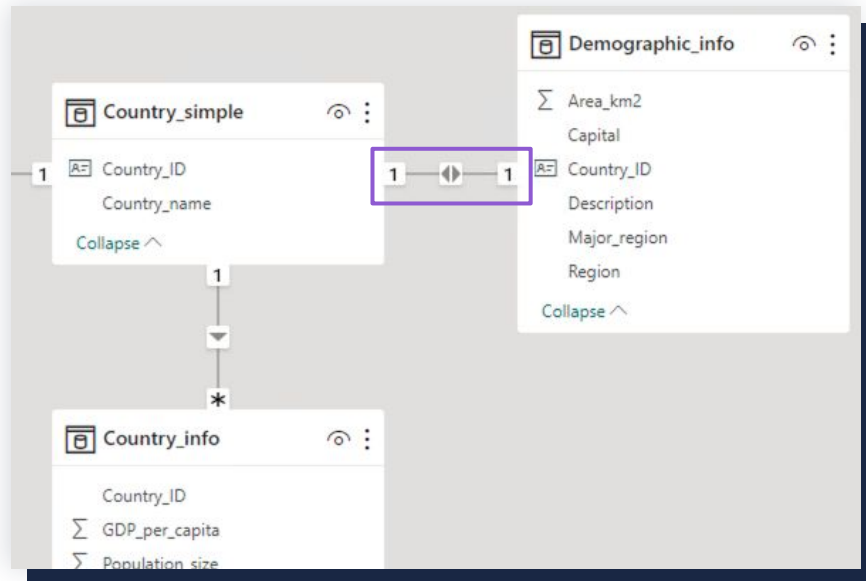
This dataset is at an **appropriate granularity** for the stories we want to tell and the **star schema** is **relatively simple**. However, we'll see how this **simple design** may **limit what we can visualise** and **how we can filter** across our tables.

One-to-one (1:1)

In a one-to-one cardinality, **each record in the first table** is associated with **one and only one record in the second table**, and vice versa.

As an example, we could move all of our geographical information per country (that doesn't change over time) to a **dedicated table called Demographic_info** that we can **enrich** with additional information.

This new table would then have a **one-to-one cardinality** with the Country_simple table because **one country is related to single description, region, area, etc.**



One-to-many (1:*)

In a one-to-many cardinality, **each record in the first table can be related to multiple records in the second table**, but each record in the second table is associated with one record in the first table.

Here we have **three one-to-many** relationships:

01. A **single country** can have **multiple records for different years and inequality indicators**, but each record in Gender_inequality is related to one specific country.
02. A **single country** can have **multiple records for different years and GDP per capita, population size**, etc. combinations, but each record in Country_info is related to one specific country.
03. A **single indicator** can have **multiple records for different years and countries**, but each record in Country_info is related to one specific indicator.

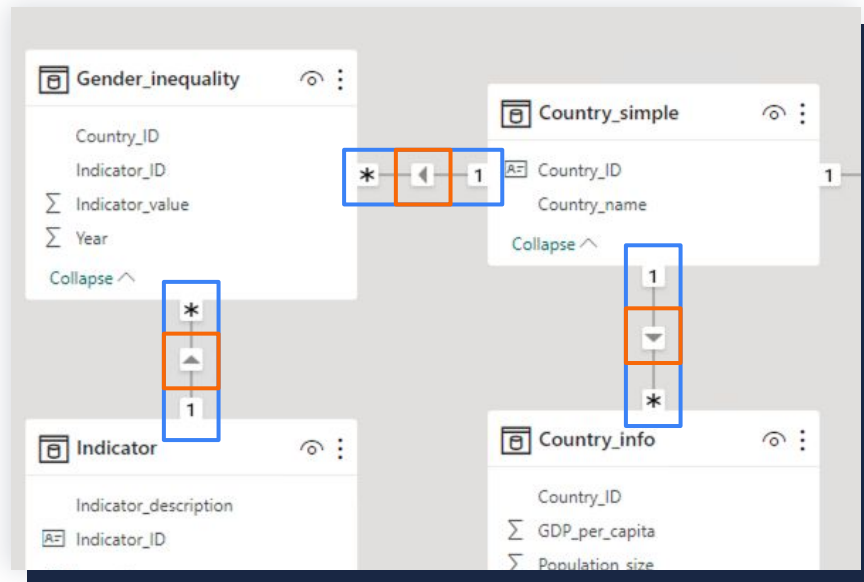


Many-to-one (*:1)

On the contrary, in a many-to-one cardinality, **each record in the first table is associated with a single record in the second table**, but each record in the second table can be linked to multiple records in the first table.

Considering the definitions of the **many-to-one** and **one-to-many** cardinalities, we see some similarities. The **difference is the direction of the relationship**.

So, while in our example all of our relationships are currently one-to-many (with a **single** direction), we **could change the directions to be bi-directional**. This would allow **filtering in both directions** (even in the direction a many-to-one cardinality would filter) but it won't change the cardinality.



Many-to-many (*:*)

In a many-to-many cardinality, **multiple records in the first table can be associated with multiple records in the second table**.

Since the Country_simple table **only includes two features**, namely Country_ID and Country_name, we could easily get rid of the table and simply move Country_name to the Country_info table.

Now we'll need a **many-to-many** cardinality since both Gender_inequality and Country_info have **multiple instances of the same country record**.



How to choose the cardinality and directionality

Understand the data

Start by **understanding the nature** of the dataset. **Identify the relationships** between tables and how they should logically relate.

Consider the story we want to tell

What is the **purpose** of the visualisations or report we're building? How do we want **visualisations to respond** to each other and what should the **interactions** look like?

Avoid unnecessary bi-directionality

While bi-directional relationships can be useful, they can make our **data model more complex and harder to manage**.

Test and iterate

It's common to **refine** relationships as we are building and testing our visualisations and reports. We always **test** our data model to **ensure filtering and calculations work** as expected.

In summary, the cardinality and directionality we choose should align with the **data structure, visualisation and reporting needs**, and the **logical flow of data** in our model.

Challenges in relationships

Creating and managing relationships in Power BI is a critical part of data modelling and reporting. Unsurprisingly, it comes with some **common challenges that need to be addressed** to **ensure data accuracy, reporting performance, and usability**.

Ambiguous paths

When there are **multiple pathways between tables**, it can create ambiguity in filter propagation.

Can lead to:

- **Unexpected filtering results** that can result in data inconsistencies or inaccuracies.
- **"Filter leakage"** where filters propagate to unintended tables or visuals, which can cause confusion and misinterpretation of data.
- **Inefficient queries** when trying to resolve ambiguous paths, which can result in longer query execution times and slower report responsiveness.

Circular dependencies

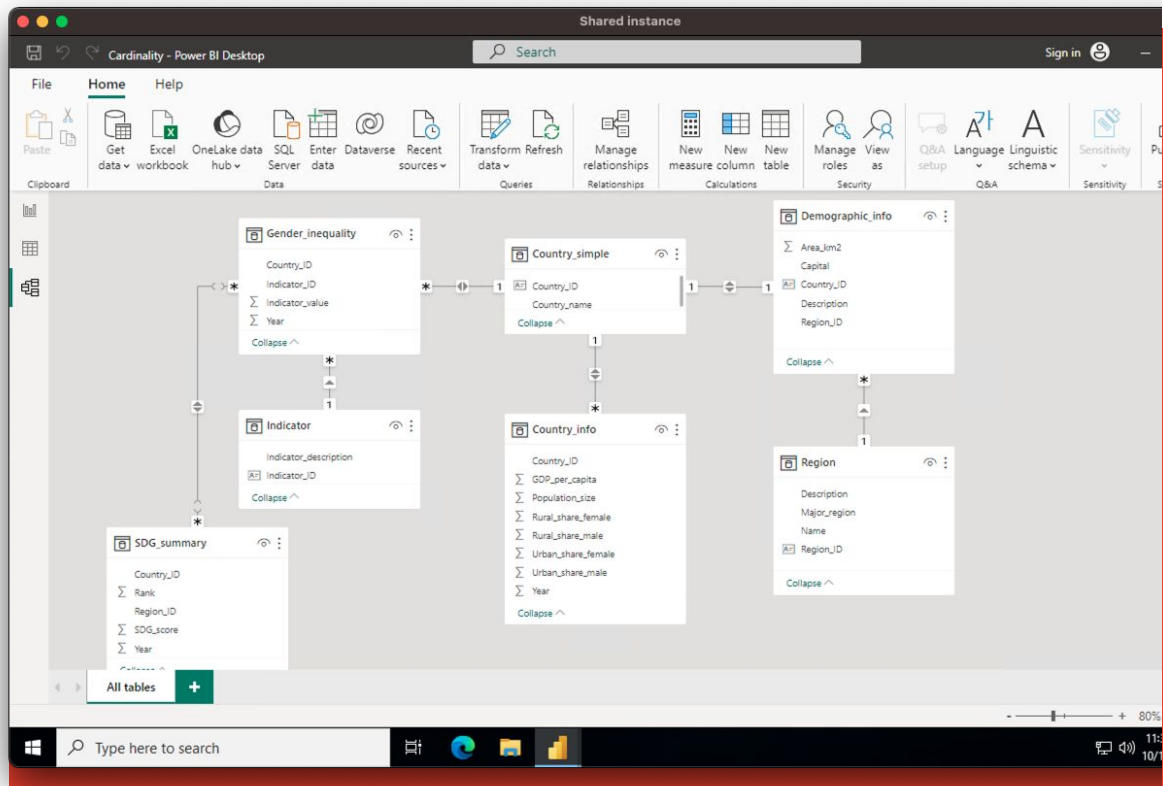
These occur **when the relationships form a loop**, and it's unclear how data should flow between tables.

Can lead to:

- **Query errors** that might prevent data from loading or displaying as expected in visualisations.
- **Performance bottlenecks** that can result in slow report rendering and user frustration.
- **Inconsistent results** where the same query or visualisation may yield different outcomes depending on the order of execution or how the relationships are resolved.

Ambiguous paths

Ambiguous paths often occur when our **data model becomes more complex**.



Ambiguous paths

Ambiguous paths often occur when our **data model becomes more complex**.

It also very often occurs when we try to activate unnecessary **many-to-many** relationships. Power BI will **always warn us on many-to-many relationships**.

Create relationship

Select tables and columns that are related.

Country_info

Country_ID	Year	GDP_per_capita	Population_size	Urban_share_male	Urban_share_female	Rural_sh
EGY	2015	\$3,933.927	97723799	21.36802588	21.7670531	
EGY	2010	\$2,922.796	87252414	21.30104001	21.71858189	
EGY	2005	\$1,331.362	79075310	21.36199618	21.66432144	

SDG_summary

Country_ID	Year	Region_ID	SDG_score	Rank
EGY	2023	NA	69.6	81
ETH	2023	EA	54.5	144
GHA	2023	WA	61.8	122

Cardinality: Many to many (*)

Cross filter direction: Both

☐ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

⚠ This relationship has cardinality Many-Many. This should only be used if it is expected that neither column (Country_ID and Country_ID) contains unique values, and that the significantly different behavior of Many-many relationships is understood. [Learn more](#)

OK Cancel

Ambiguous paths

Ambiguous paths often occur when our **data model becomes more complex**.

It also very often occurs when we try to activate unnecessary **many-to-many** relationships. Power BI will **always warn us on many-to-many relationships**.

Luckily, Power BI will most often **not allow us to make a relationship active** that could potentially cause ambiguous paths.

Create relationship

Select tables and columns that are related.

Country_info

Country_ID	Year	GDP_per_capita	Population_size	Urban_share_male	Urban_share_female	Rural_sh
EGY	2015	\$3,933.927	97723799	21.36802588	21.7670531	
EGY	2010	\$2,922.796	87252414	21.30104001	21.71858189	
EGY	2005	\$1,331.362	79075310	21.36199618	21.66432144	

SDG_summary

Country_ID	Year	Region_ID	SDG_score	Rank
EGY	2023	NA	69.6	81
ETH	2023	EA	54.5	144
GHA	2023	WA	61.8	122

Cardinality: Many to many (":*)

Cross filter direction: Both

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Warning: You can't create a direct active relationship between Country_info and SDG_summary because that would introduce ambiguity between tables SDG_summary and Country_simple. To make this relationship active, deactivate or delete one of the relationships between SDG_summary and Country_simple first.

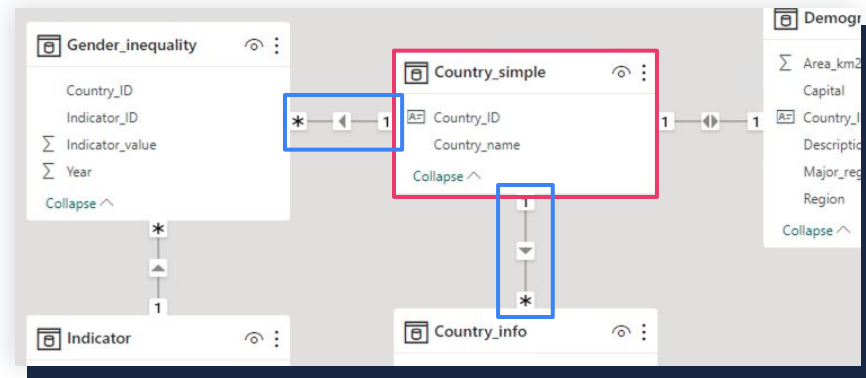
How to avoid challenging relationships

We've seen that many-to-many relationships often contribute to the common challenges we see in Power BI.

A simple **alternative to many-to-many relationships** is to use a **bridge table**, also known as a **junction table** or **association table**.

Think back to the **many-to-many** example we had and compare that to the **one-to-many** example.

So, we've technically already seen an example of a **bridge table** in action. We've **replaced the many-to-many relationship with two one-to-many** relationships by introducing the Country_simple table.



Tips for relationships in Power BI

Keep relationships simple

Try to keep to **one-to-many** relationships with a single directionality. They are **easy to interpret** and avoid over-complicating the data model.

While one-to-one and many-to-many relationships are useful in very specific situations, they may cause ambiguous paths and circular dependencies. Try to rather **use bridge tables** in many-to-many scenarios.

Iterate on the data structure

Restructure the data by moving features to a different table or **pivoting the data** if the interactivity between visualisations is not as expected.

Change the granularity of the dataset. **Simplify** the dataset if the data model and visualisations become too complex, or **enrich** the dataset if more detail is required to the data story.